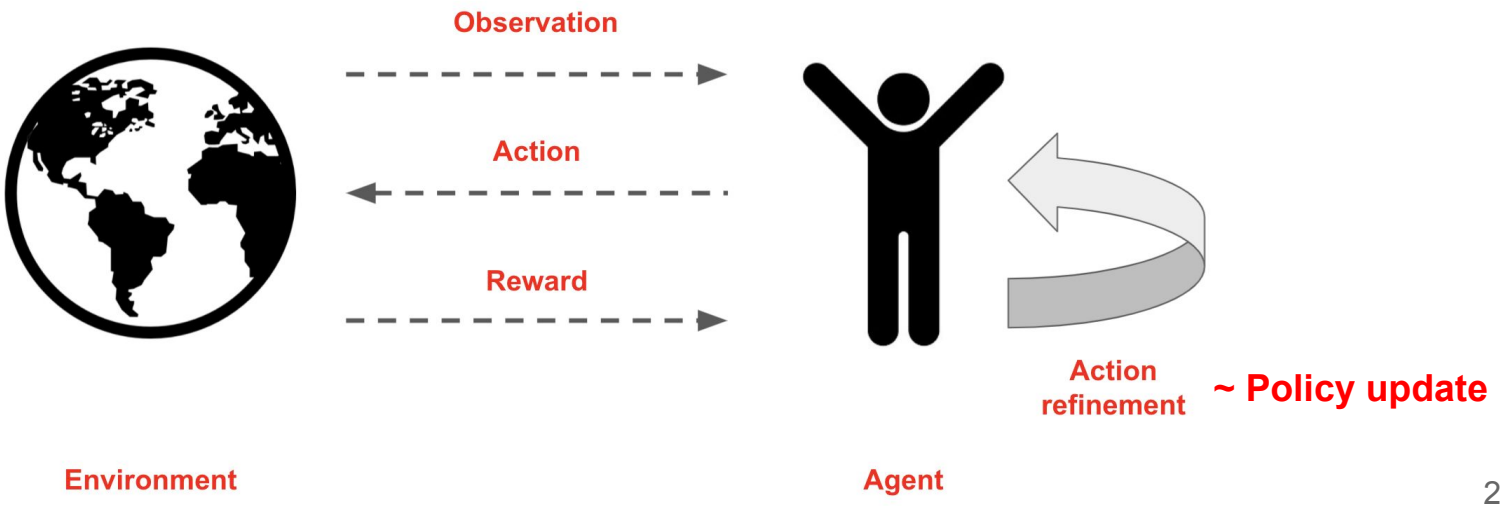
Markov Decision Processes

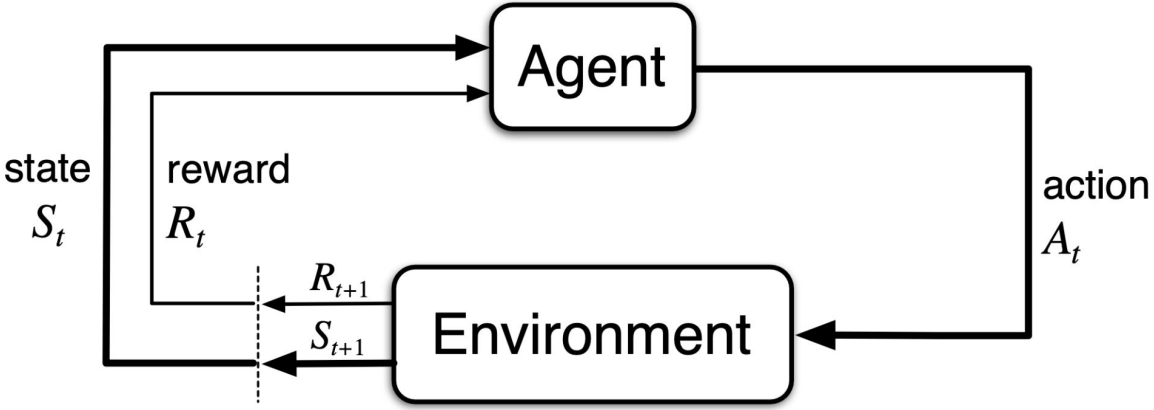
Genya Ishigaki

Review: Reinforcement Learning Framework

- An agent takes an action given an observation
- The effectiveness of the action is indicated by a reward value
- An agent updates its policy

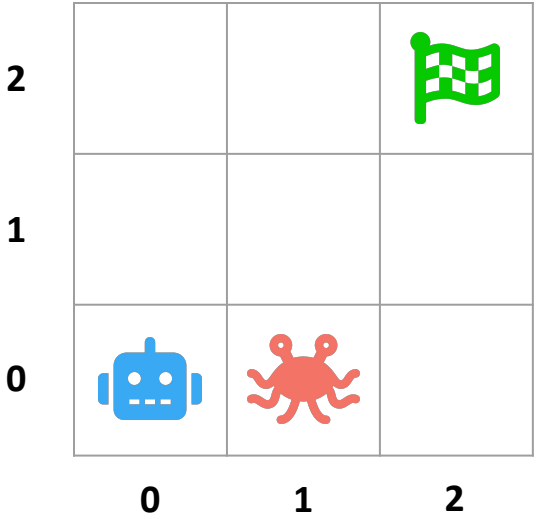


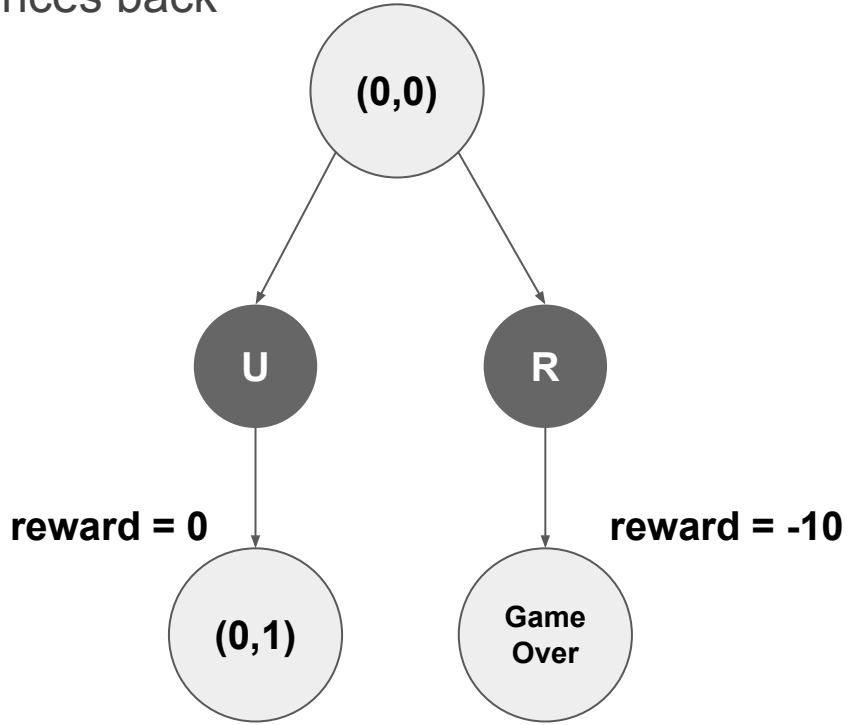
The Agent-Environment Interaction



Markov Decision Process (MDP): An Example

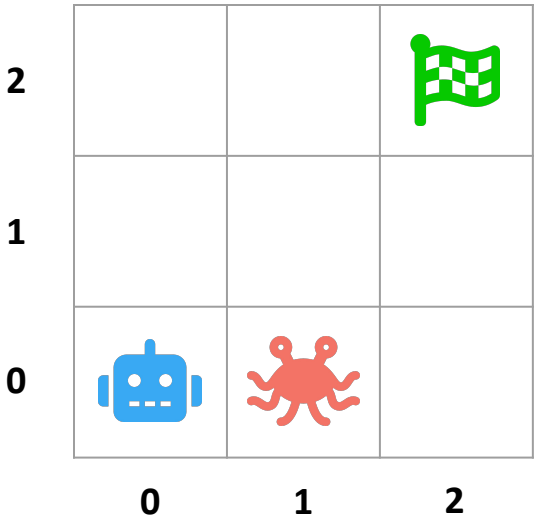
- Action (R, L, U, D)
- When an agent hits a wall, it bounces back





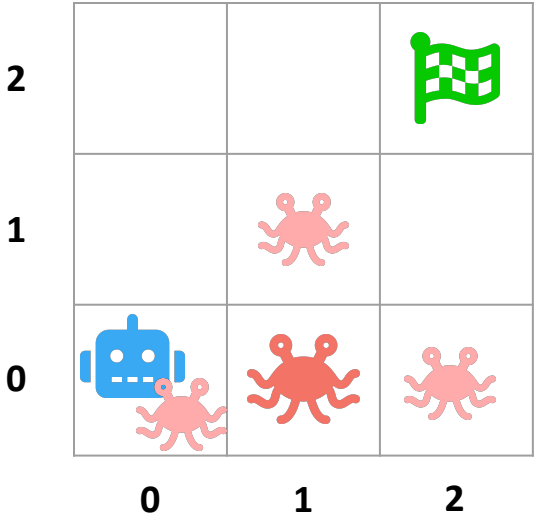
Milestone Questions

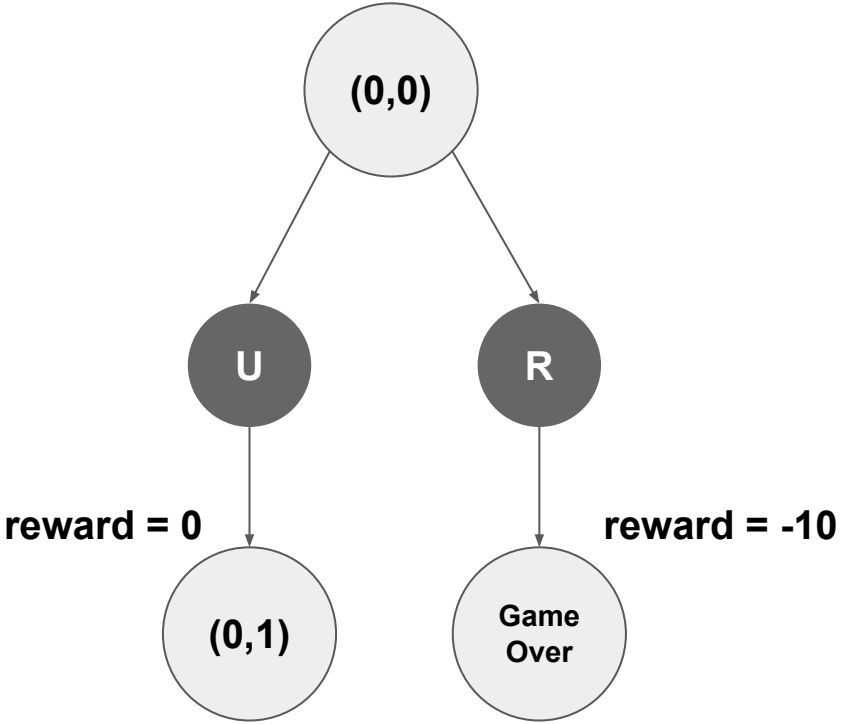
- How many states do we have in the MDP?
- Draw a part of the MDP starting from state (1,1).



Markov Decision Process (MDP): An Example

- What happens if the monster takes one of actions (R, L, U, D=Stay) uniformly randomly?





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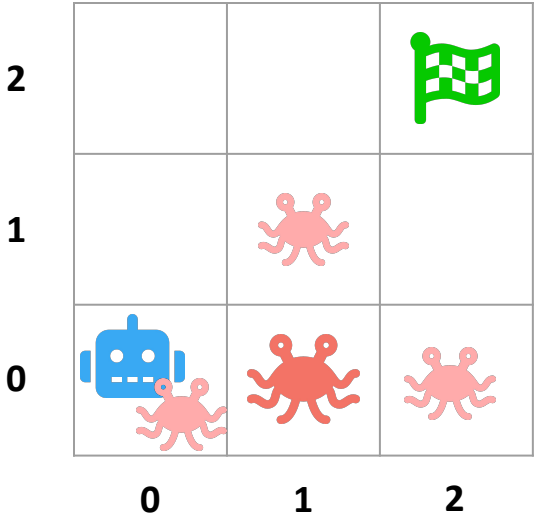
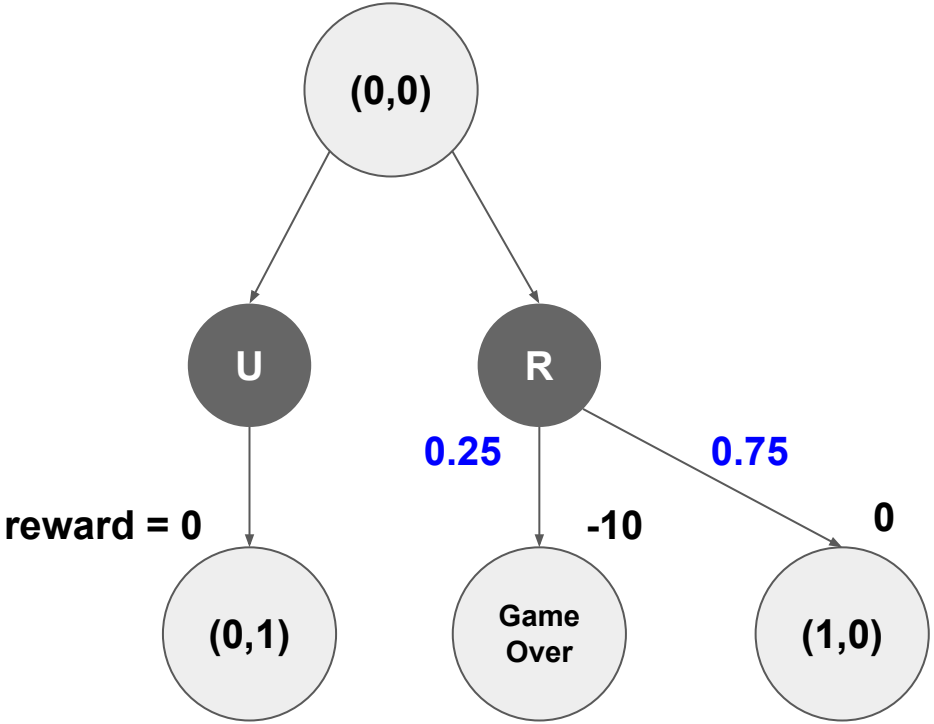




Markov Decision Process (MDP)

A Markov Decision Process is a tuple <S, A, P, R, γ>

- S: A finite set of states
- A: A finite set of actions
- P: A state transition probability matrix

$$P\left[s' \mid s, a\right]$$

- R: A reward function

$$R\left(s, a, s'\right)$$

- γ: A discount factor in [0, 1]

A Policy

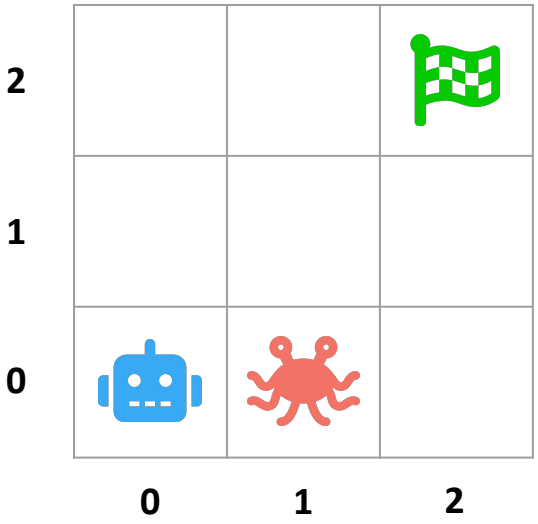
- A policy is a distribution over actions given states

$$\pi(a \mid s) = P[a \mid s]$$

- A policy fully defines the behavior of an agent
- MDP policies depend on the current state (not the history or a sequence of actions)

Milestone Questions

- Assume you decided to use a uniformly random policy from every state
- Formally define the policy function $\pi(a \mid s)$



Milestone Questions

- Both a policy and a transition probability decides the state transition of an agent
- What is the difference between the two?

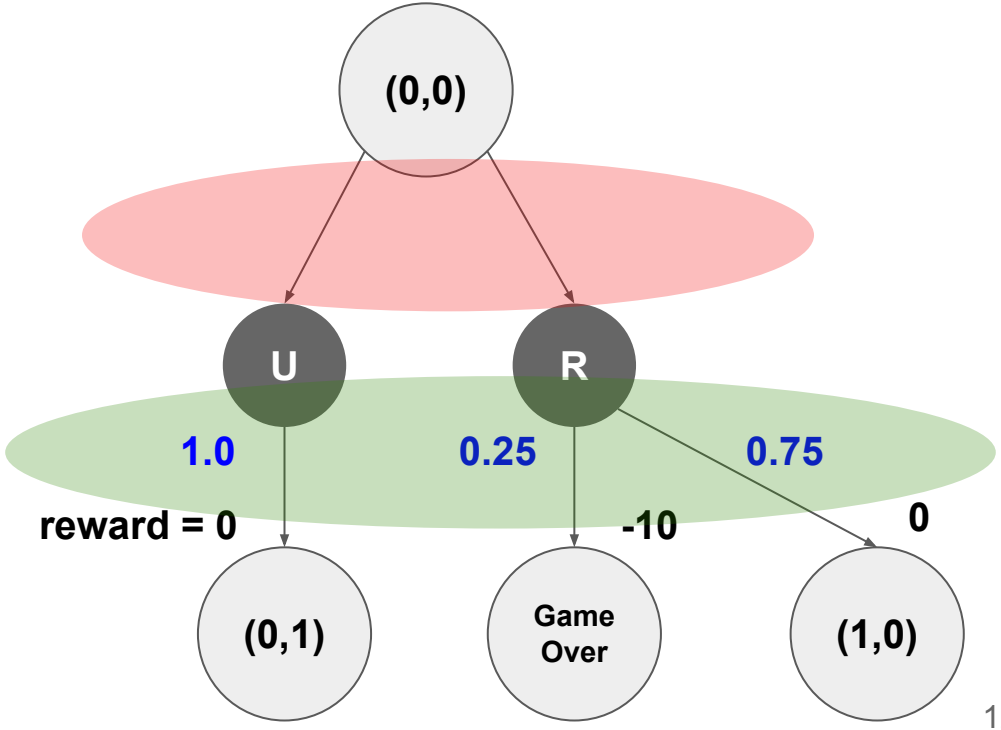
Policy vs Transition Probability

Policy defines which action to take

- Intentional
- You have control over it
- A better policy gives better rewards

Transition decides where to go

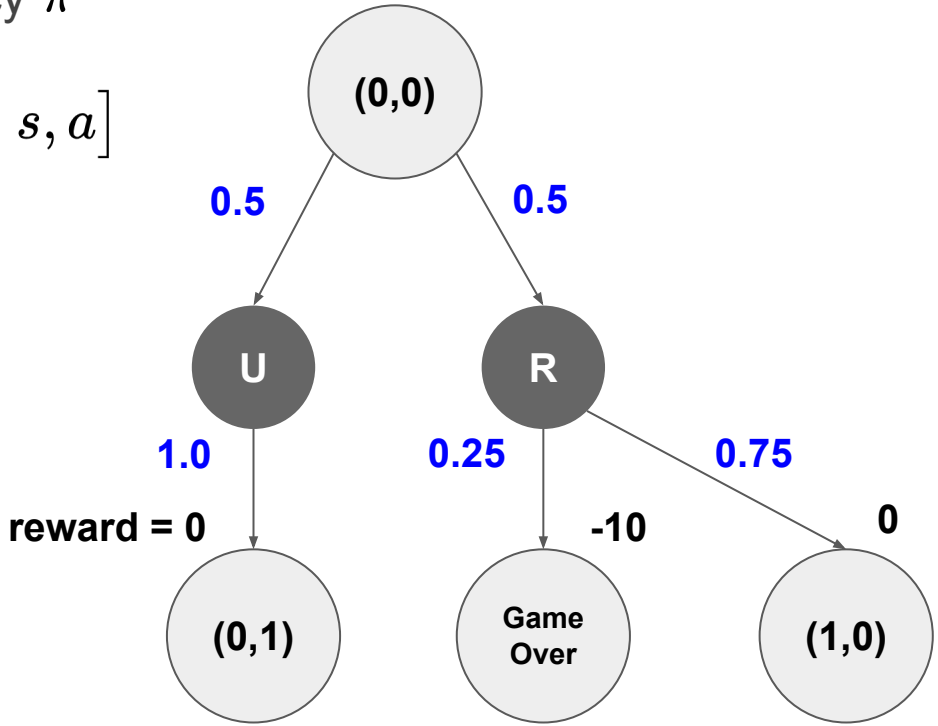
- Given in the environment
- You do NOT have control
- A better policy should be selected guessing this probability



The Probability of Moving from s to s'

- How can we compute the probability of moving from s to s', given a policy π

$$P_\pi[s' \mid s] = \sum_{a \in A} \pi(a \mid s) P[s' \mid s, a]$$

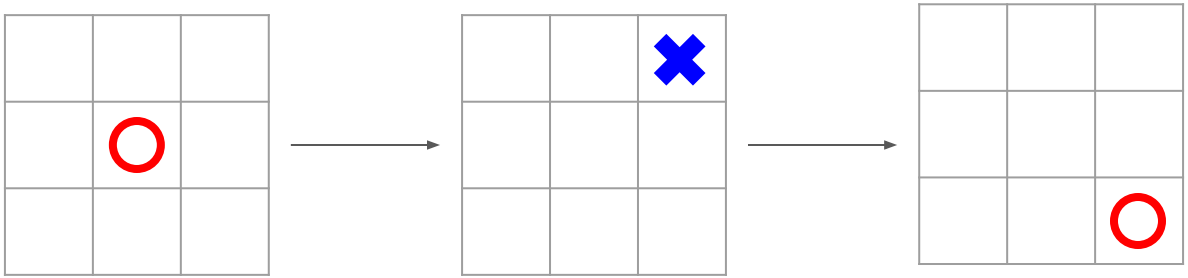


The Markov Property

- The future states of a stochastic process are influenced only by the present, not the past, meaning that the past can be disregarded once the present is known

$$P[S_{t+1} \mid S_t] = P[S_{t+1} \mid S_1, \ldots, S_t]$$

- Think about the states of Tic-Tac-Toe
- Is this Markovian?



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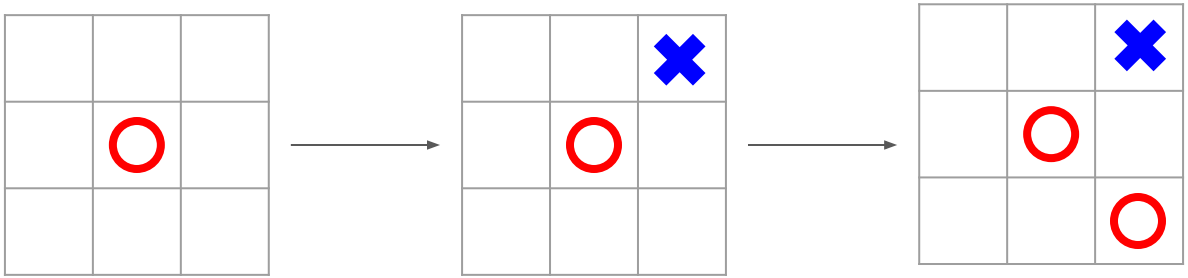


The Reward Hypothesis

- All of what we mean by goals and purposes can be well thought of as the maximization of the expected value of the cumulative sum of a received scalar signal (called reward)
- Utility theory
  - Explaining individuals' choices and measures their level of satisfaction from consuming a good or service

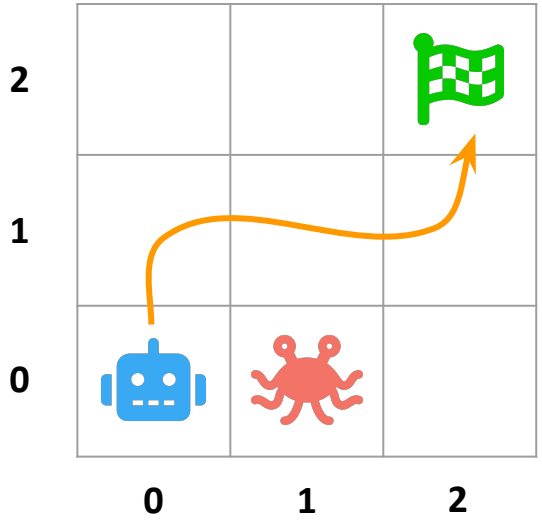
Milestone Questions

Design a reward function for each of the following tasks

- Making a robot learn to find and collect empty soda cans for recycling
- Making a robot learn how to escape from a maze as quickly as possible

Returns and Episodes

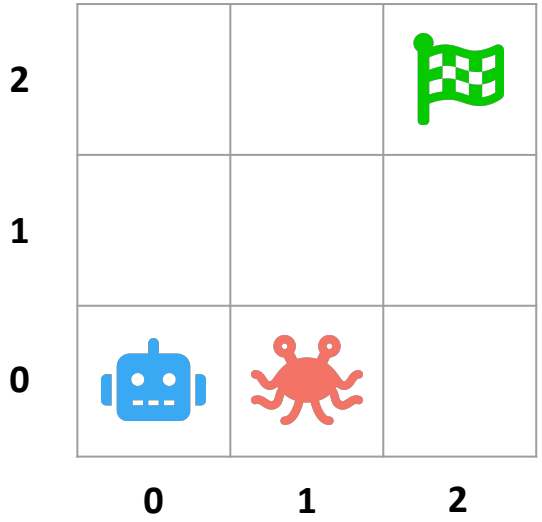
- A reward
  - An immediate feedback signal for an action of the agent
- Our objective
  - To maximize the cumulative reward it receives in the long run
- An episode
  - A sequence of state s, action a, and reward r
  - e.g. Represent the episode of the arrow
    - Each step (except for the goal state) -1
    - Goal state +10



Returns and Episodes

- (0,0), U, -1, (0,1), R, -1, (1,1), R, -1, (2,1), U, 10, (Goal)
- What is the return G of this episode?
  - G = -1 + -1 + -1 + 10
- Formally, a return from State St is

$$G_t \doteq R_{t+1} + R_{t+2} + R_{t+3} + \cdots + R_T$$



Milestone Questions

- What is a difference between a reward and a return

Episodic vs. Continuing Tasks

- Each episode ends in a special state called the terminal state:
  - The task is called “episodic”
  - Finite length of actions always bring the agent to the end of a trial
- The agent-environment interaction does not break naturally into identifiable episodes, but goes on continually without limit
  - The task is called “continuing”
  - Infinite length of actions

Milestone Questions

Come up with examples of episodic and continuing tasks

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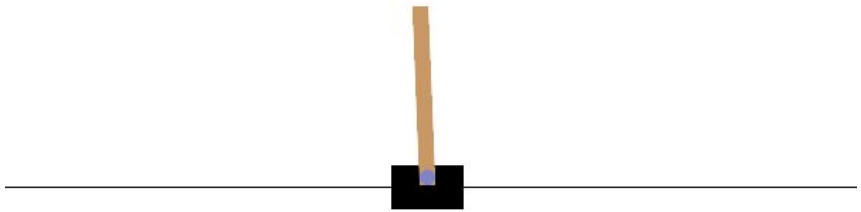
Come up with examples of episodic and continuing tasks

E.g., Cart pole



A Discounting Factor

- A discount factor determines the present value of future rewards

$$G_t \doteq R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \cdots = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$$

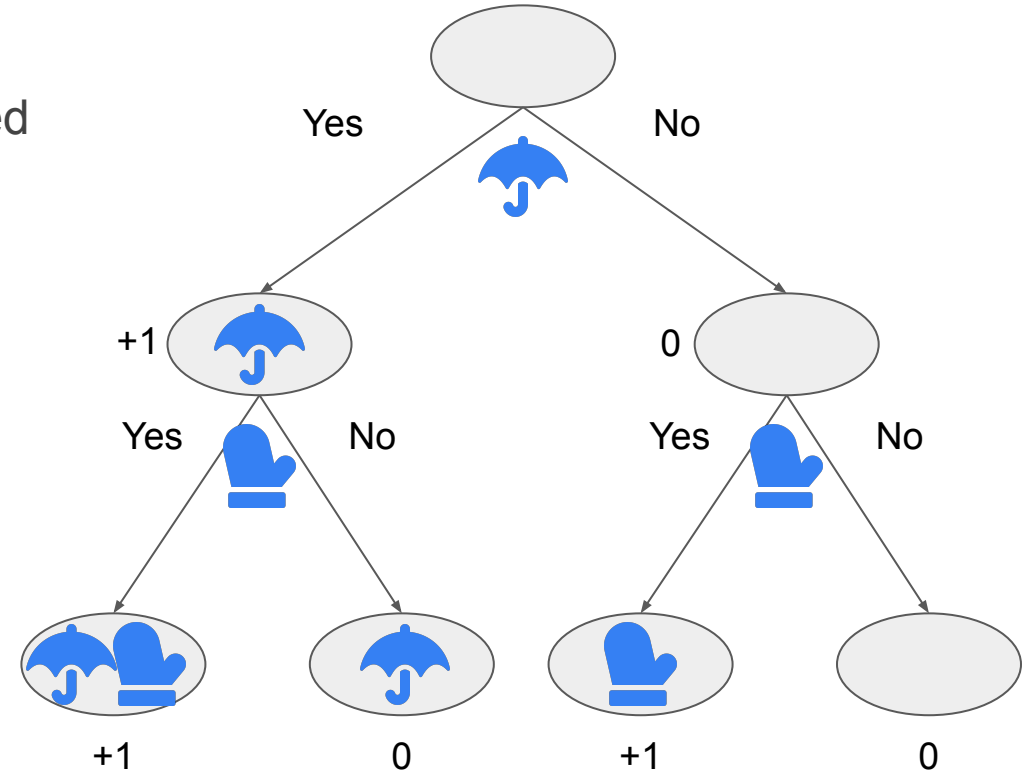
$$0 \leq \gamma \leq 1$$

- If γ = 0, the agent is myopic
- If γ is close to 1, the agent is far-sighted

Part II

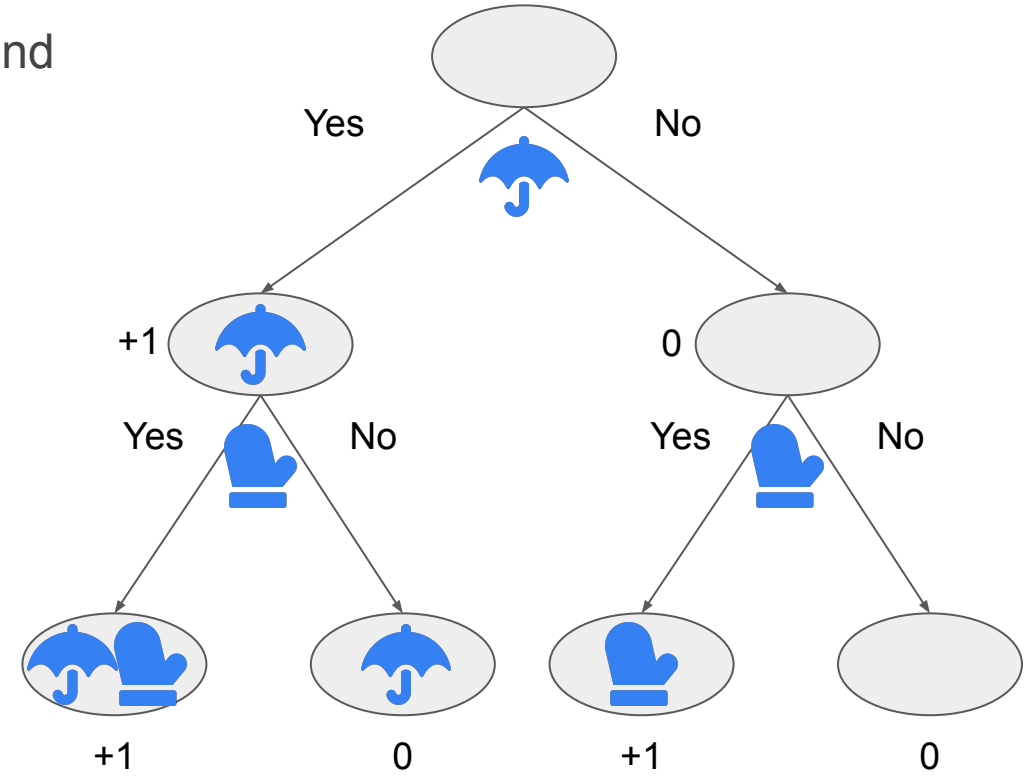
The Returns

- A return represents a long-term feedback for a sequence selected by your actions
- E.g. On a cold rainy day…



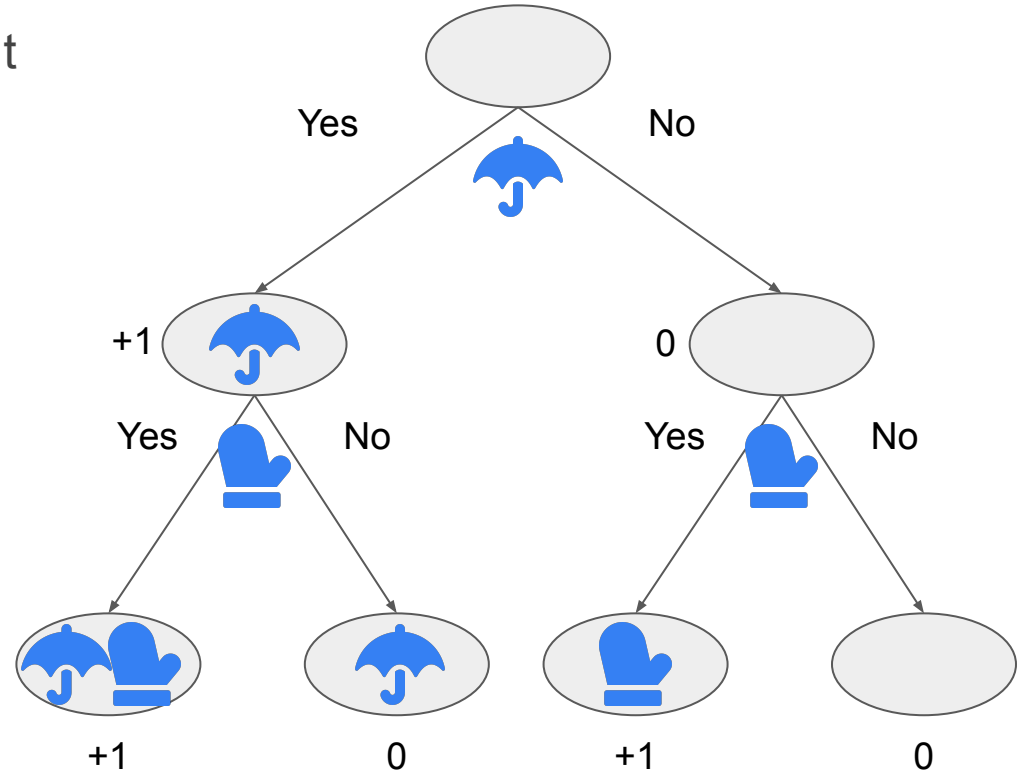
The Returns

- Alice always bring an umbrella and wear mittens.
  - How do you represent this Alice's policy?



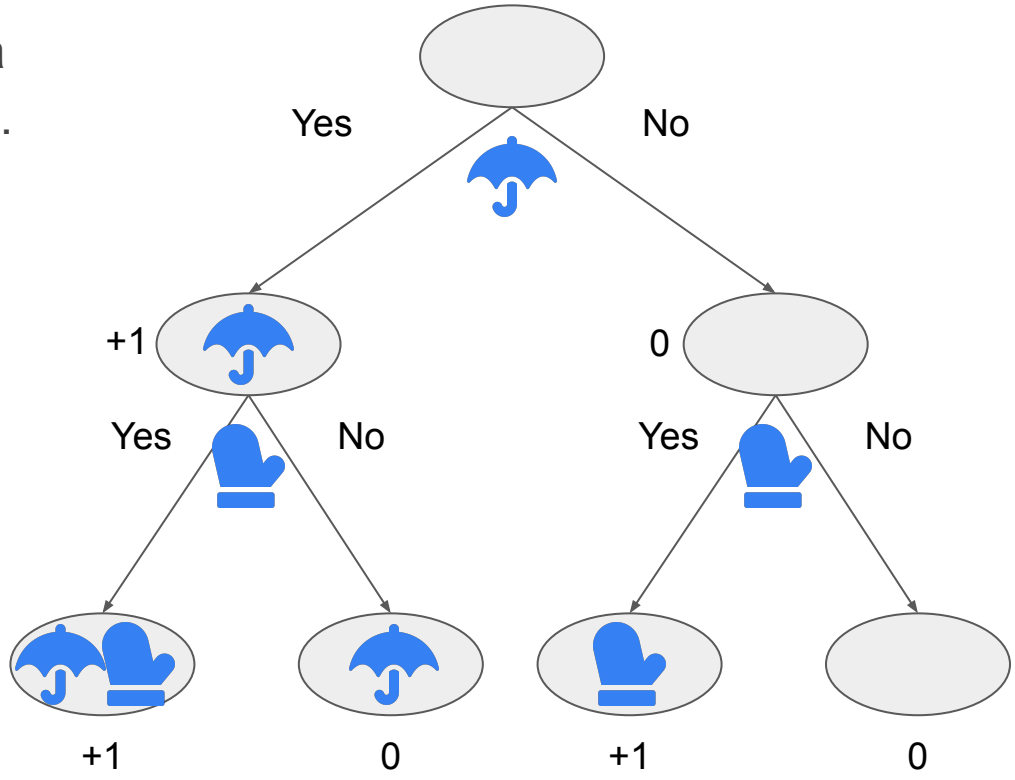
The Returns

- Bob always bring an umbrella but does not like wearing mittens.
  - How do you represent this Bob's policy?



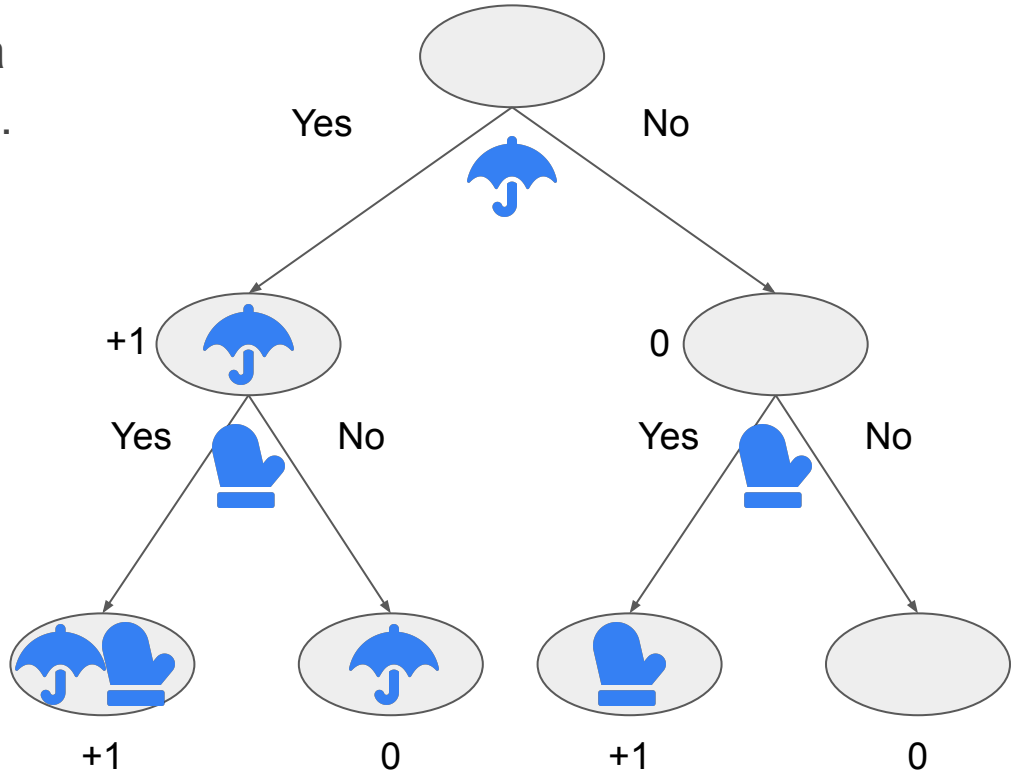
Milestone Questions

- Cassey always bring an umbrella but wear mittens 50% of the time.
  - How do you represent this Cassey's policy?



Milestone Questions

- Cassey always bring an umbrella but wear mittens 50% of the time.
  - How do you represent this Cassey's policy?
  - Over time, Cassey will take two types of sequences with the equal probability
    - Yes -> No
    - Yes -> Yes

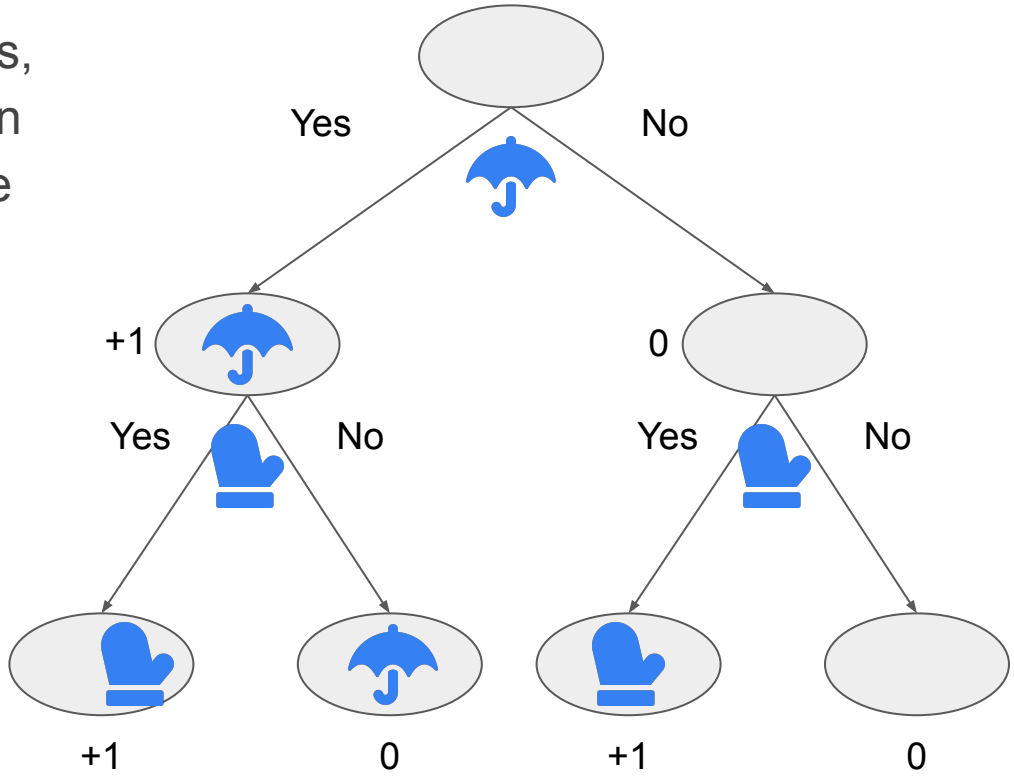


The State Value Function

- When you want to evaluate states, it is NOT enough to use the return from a single trial (Remember the Cassey's case)
- We need to assess the average return possible from the state

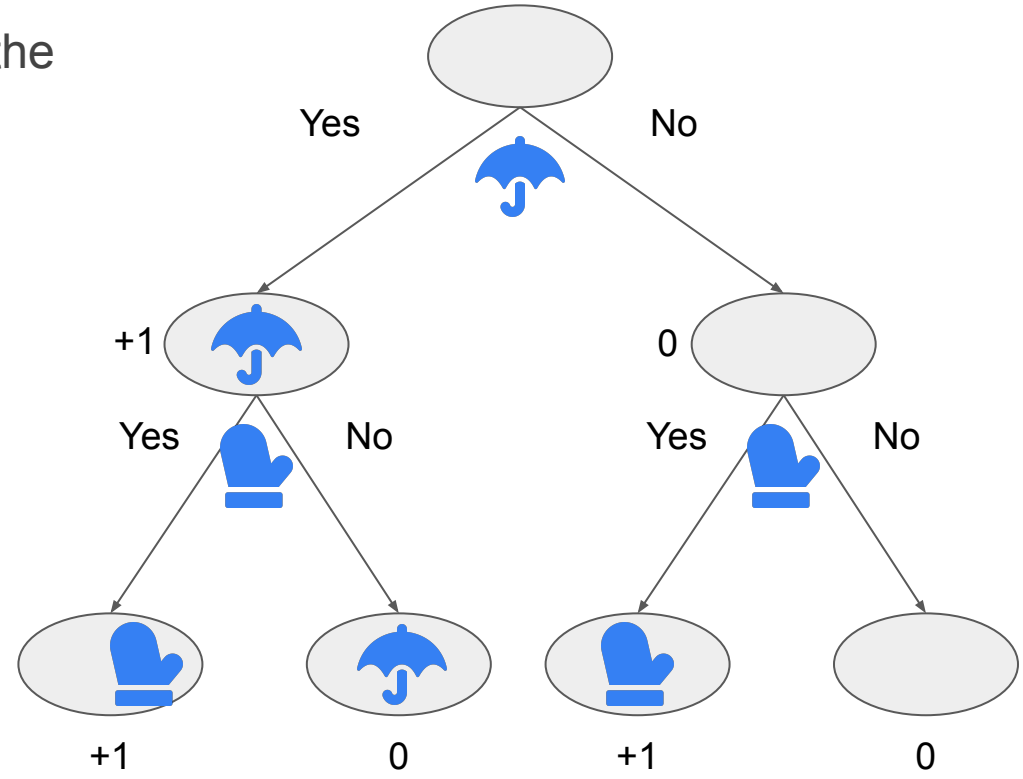
$$v_\pi(s) \doteq \mathbb{E}_\pi[G_t \mid S_t = s]$$

$$= \mathbb{E}_\pi\left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \;\middle|\; S_t = s\right]$$



The State Value Function

- What is the effect of a policy on the value of state?



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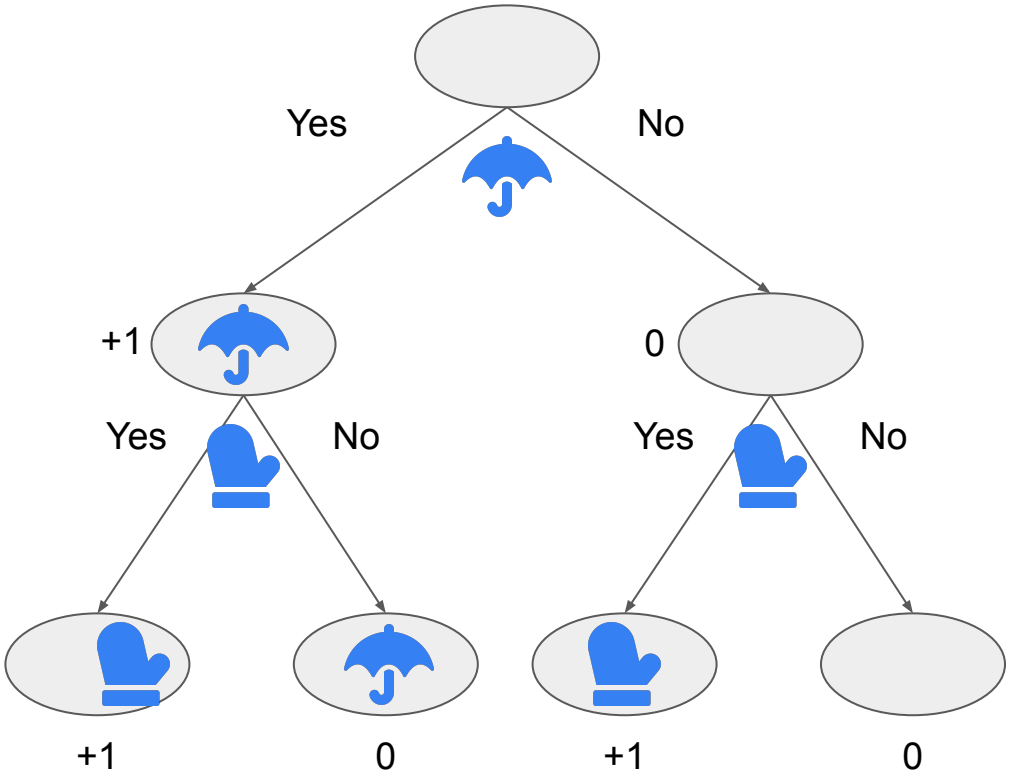
$$= \mathbb{E}_\pi\left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \;\middle|\; S_t = s\right]$$

The Action Value Function

- Similar to the value of state, we can define the average return from a state taking an action

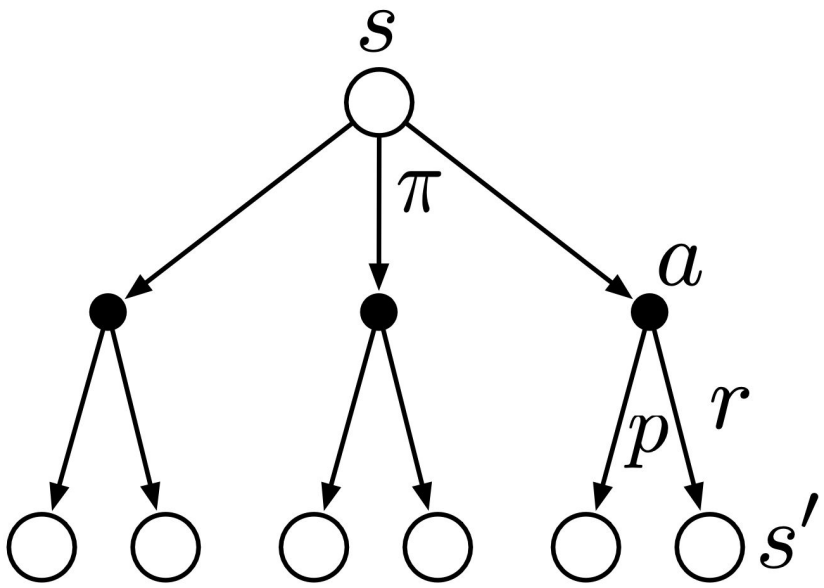
$$q_\pi(s,a) \doteq \mathbb{E}_\pi[G_t \mid S_t = s, A_t = a]$$

$$= \mathbb{E}_\pi\left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \;\middle|\; S_t = s, A_t = a\right]$$



The Recursive Relationship of Value Functions

$$
\begin{aligned}
v_\pi(s) &\doteq \mathbb{E}_\pi[G_t \mid S_t = s] \\
&= \mathbb{E}_\pi[R_{t+1} + \gamma G_{t+1} \mid S_t = s]
\end{aligned}
$$



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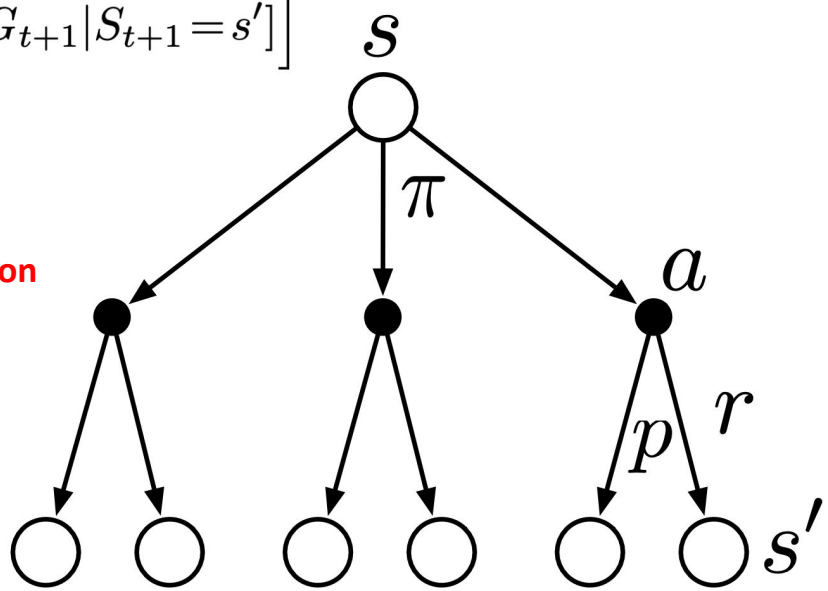
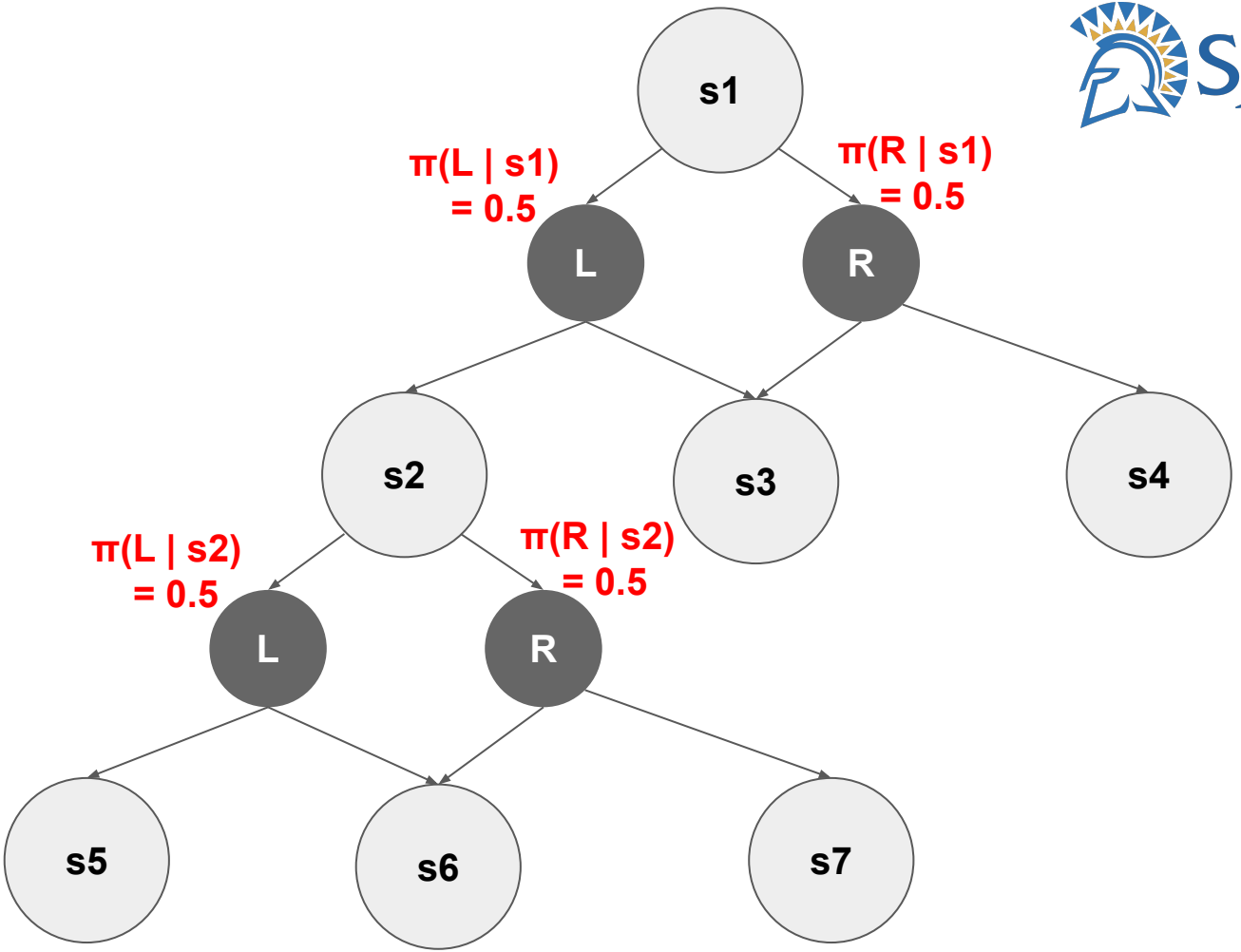
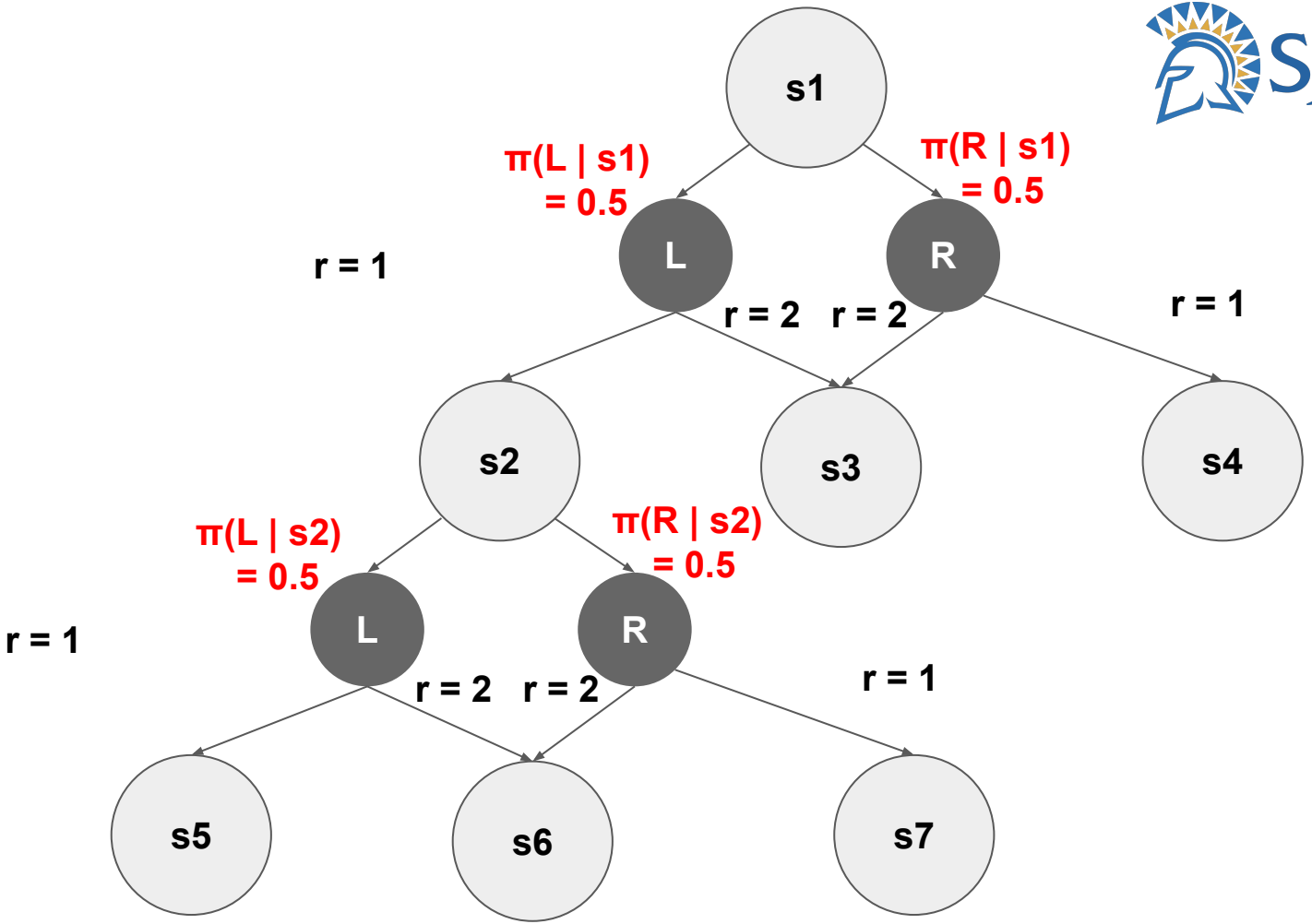
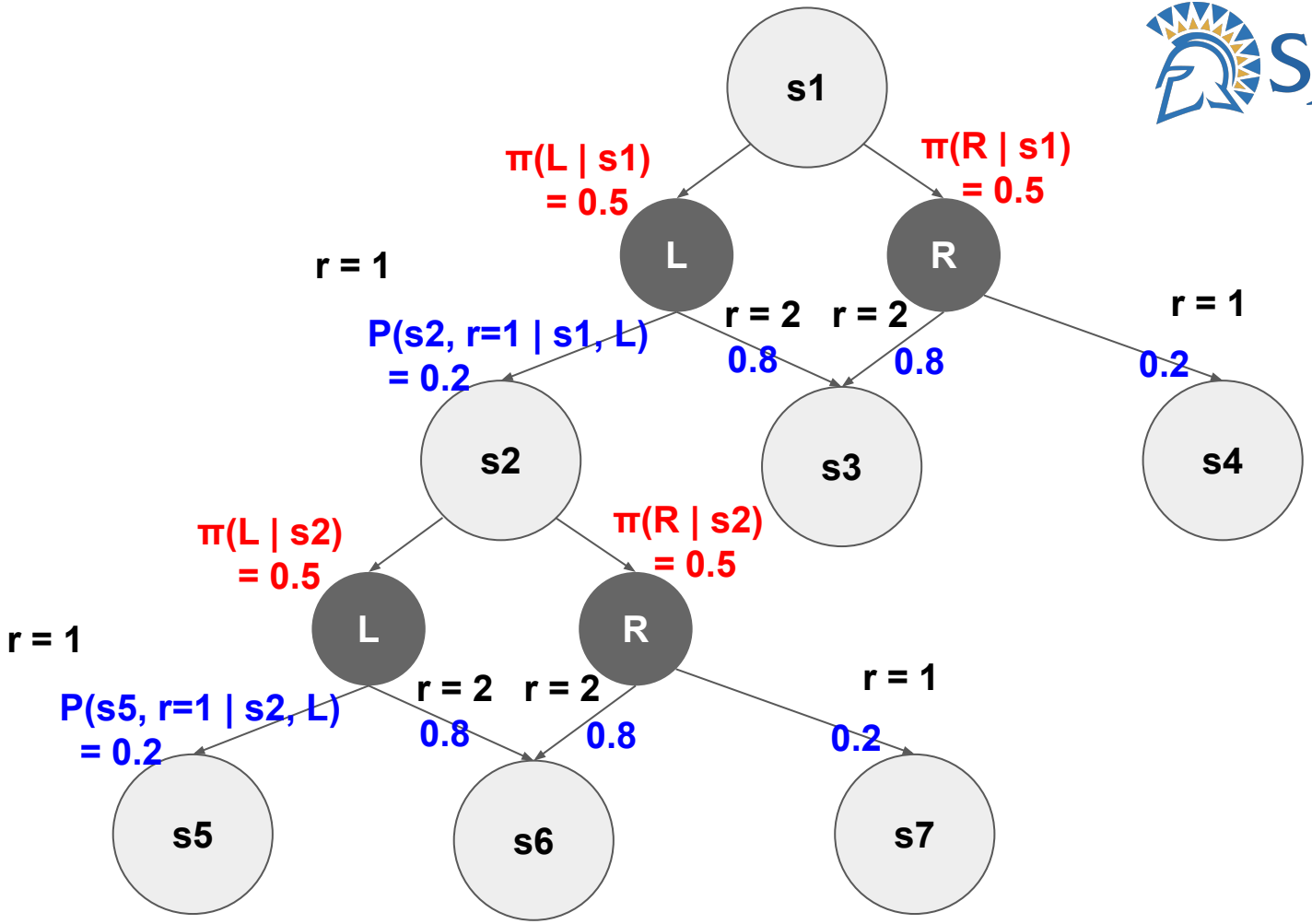
&= \mathbb{E}_\pi[R_{t+1} + \gamma G_{t+1} \mid S_t = s] \\
&= \sum_a \pi(a|s) \sum_{s'} \sum_r p(s', r \mid s, a) \Big[r + \gamma \mathbb{E}_\pi[G_{t+1} | S_{t+1} = s'] \Big] \\
&= \sum_a \pi(a|s) \sum_{s', r} p(s', r \mid s, a) \Big[r + \gamma v_\pi(s') \Big]
\end{aligned}
$$

The Bellman Expectation Equation for the value function




s1
π(L | s1)
= 0.5
π(R | s1)
= 0.5
L
R
s2
s3
s4
π(L | s2)
= 0.5
π(R | s2)
= 0.5
L
R
s5
s6
s7


s1
π(L | s1)
= 0.5
π(R | s1)
= 0.5
L
R
r = 1
r = 2
r = 2
r = 1
s2
s3
s4
π(L | s2)
= 0.5
π(R | s2)
= 0.5
L
R
r = 1
r = 2
r = 2
r = 1
s5
s6
s7


s1
π(L | s1)
= 0.5
π(R | s1)
= 0.5
L
R
r = 1
r = 2
r = 2
r = 1
P(s2, r=1 | s1, L)
= 0.2
0.8
0.8
0.2
s2
s3
s4
π(L | s2)
= 0.5
π(R | s2)
= 0.5
L
R
r = 1
r = 2
r = 2
r = 1
P(s5, r=1 | s2, L)
= 0.2
0.8
0.8
0.2
s5
s6
s7

The Bellman Expectation Equation

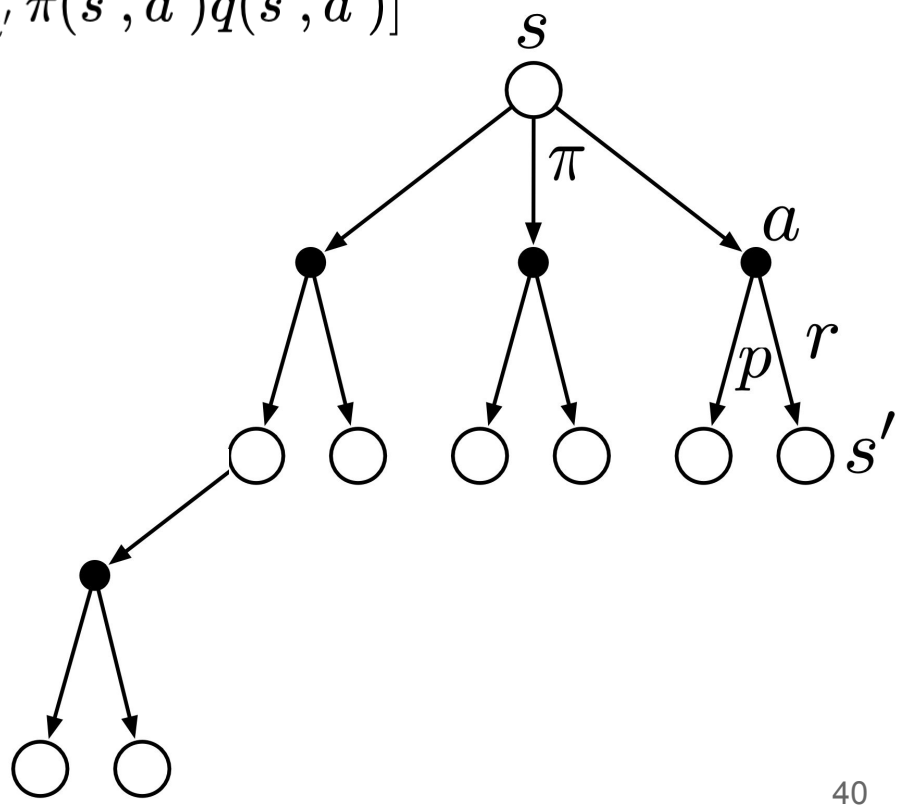
$$v_\pi(s) \doteq \sum_a \pi(a|s) \sum_{s',r} p(s',r|s,a)\Big[r + \gamma v_\pi(s')\Big]$$

- Averages over all the possibilities, weighting each by its probability of occurring
- The start state must equal the (discounted) value of the expected next state, plus the reward expected along the way
- Forms the basis of a number of ways to compute, approximate, and learn the value function

The Bellman Expectation Equation

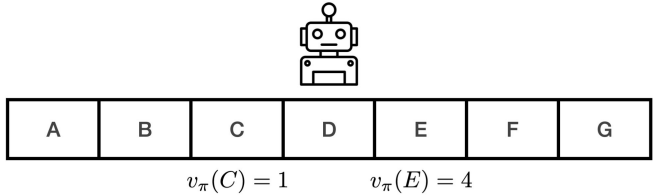
$$q_\pi(s,a) \doteq \sum_{s',\,r} p(s',\,r \mid s,a)[r + \gamma \sum_{a'} \pi(s',a')q(s',a')]$$

- Averages over all the possibilities, weighting each by its probability of occurring from a state-action pair



Milestone Questions

- A robot is in a cell (State)
- I can move the robot to right or left (Two possible actions)
- Now, from my experience, I decided to choose RIGHT in cell D 60% of the time (Policy)
- BUT the robot is malfunctioning and moves to the opposite direction with 20% (Transition probability)
- Compute the state value of D (Use a discount factor of 1.0)



$v_\pi(C) = 1$ $v_\pi(E) = 4$

$$R_D^{right} = 3$$
$$R_D^{left} = 3$$

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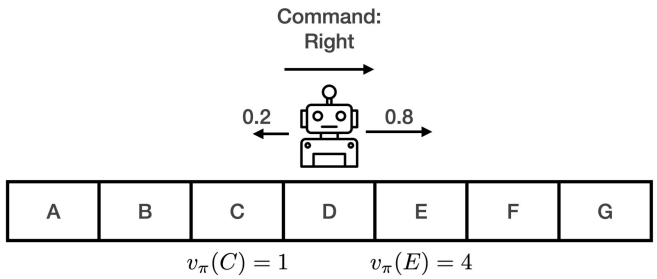
$$q_\pi(D, \text{right}) = \mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{ss'}^a v_\pi(s')$$



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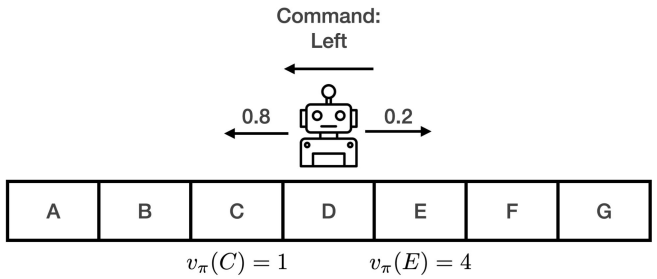
- Now, from my experience, I decided to choose RIGHT in cell D 60% of the time (Policy)
- BUT the robot is malfunctioning and moves to the opposite direction with 20% (Transition probability)

$$q_\pi(D, \text{left}) = \mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{ss'}^a v_\pi(s')$$



$v_\pi(C) = 1$ $\quad$ $v_\pi(E) = 4$

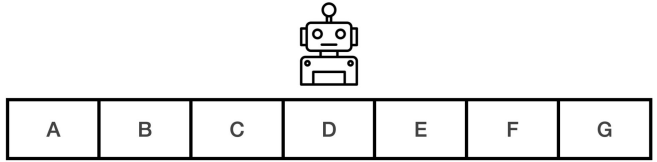
$R_D^{right} = 3$

$R_D^{left} = 3$

Milestone Questions

- A robot is in a cell (State)
- I can move the robot to right or left (Two possible actions)
- Now, from my experience, I decided to choose RIGHT in cell D 60% of the time (Policy)
- BUT the robot is malfunctioning and moves to the opposite direction with 20% (Transition probability)

$$v_\pi(D) = \sum_{a \in \mathcal{A}} \pi(a \mid s) q_\pi(s, a)$$



Optimal Policies and Optimal Value Functions

- Solving a reinforcement learning task
- = finding a policy that achieves a lot of reward over the long run
- The Bellman expectation equation (the values of states) depends on a policy
- We want to find an optimal policy
  - A policy π is said better than a policy π'
  - if its expected return is greater than or equal to that of π''s expected return

$$\pi \geq \pi' \text{ if and only if } v_\pi(s) \geq v_{\pi'}(s) \quad \text{for all states}$$

The Optimal Value Functions

- The optimal state value function (for any state s)

$$v_*(s) \doteq \max_\pi v_\pi(s)$$

- The optimal action value function (for any state-action pair s, a)

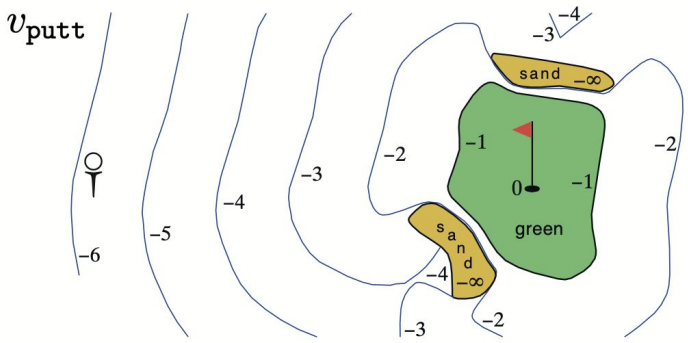
$$q_*(s,a) \doteq \max_\pi q_\pi(s,a)$$

- For the state-action pair (s,a), this function gives the expected return for taking action a in state s and thereafter following an optimal policy

$$q_*(s,a) = \mathbb{E}[R_{t+1} + \gamma v_*(S_{t+1}) \mid S_t = s, A_t = a]$$

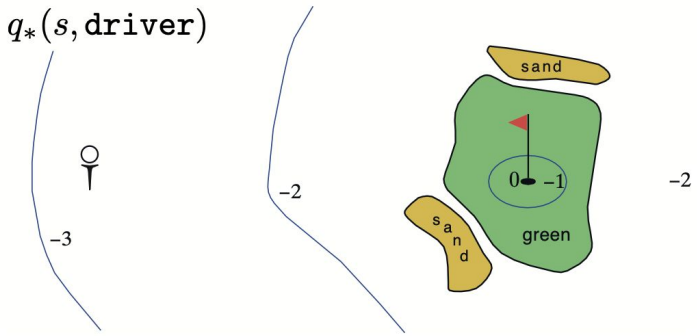
An Example

- Playing a hole of golf as a reinforcement learning task
  - A penalty (negative reward) of 1 for each stroke until we hit the ball into the hole
  - The state is the location of the ball
  - The value of a state is the negative of the number of strokes to the hole from that location
  - Our actions are which club we select - **putter or driver** (assuming we can perform a nice swing with a selected club)



An Example

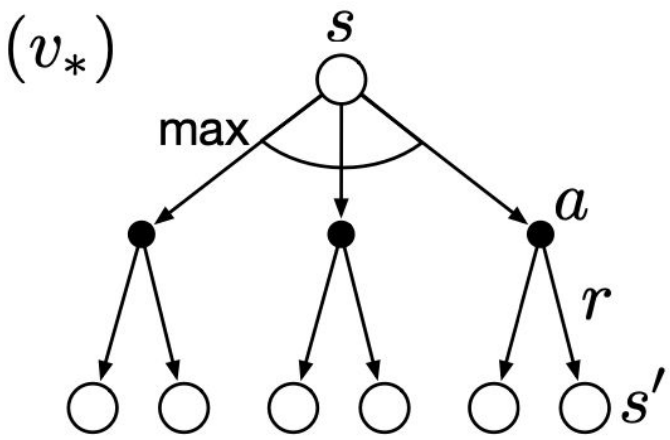
- The optimal action-value function evaluate the state-action pair (s, driver), assuming that
  - we first play a stroke with the driver and
  - afterward select either the driver or the putter, whichever is better
- The optimal action-value function gives the values after committing to a particular first action, in this case, to the driver, but afterward using whichever actions are best



The Bellman Optimality Equation

- Expresses the fact that the value of a state under an optimal policy must equal the expected return for the best action from that state

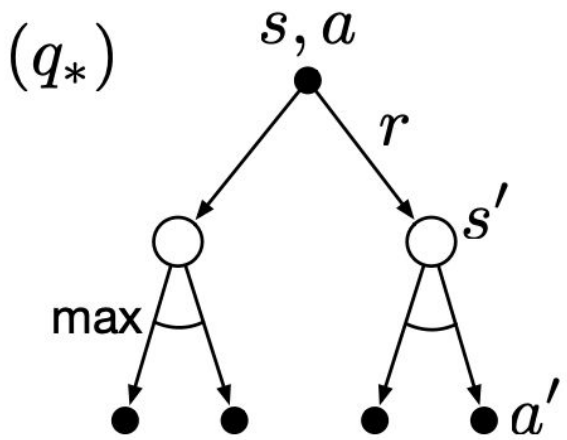
$$
\begin{aligned}
v_*(s) &= \max_{a \in \mathcal{A}(s)} q_{\pi_*}(s,a) \\
&= \max_a \mathbb{E}_{\pi_*}[G_t \mid S_t = s, A_t = a] \\
&= \max_a \mathbb{E}_{\pi_*}[R_{t+1} + \gamma G_{t+1} \mid S_t = s, A_t = a] \\
&= \max_a \mathbb{E}[R_{t+1} + \gamma v_*(S_{t+1}) \mid S_t = s, A_t = a] \\
&= \max_a \sum_{s',r} p(s',r \mid s,a)\big[r + \gamma v_*(s')\big].
\end{aligned}
$$



The Bellman Optimality Equation

- Expresses the fact that the value of a state under an optimal policy must equal the expected return for the best action from that state

$$
\begin{aligned}
q_*(s,a) &= \mathbb{E}\Big[R_{t+1} + \gamma \max_{a'} q_*(S_{t+1}, a') \;\Big|\; S_t = s, A_t = a\Big] \\
&= \sum_{s',r} p(s',r \mid s,a)\Big[r + \gamma \max_{a'} q_*(s',a')\Big].
\end{aligned}
$$



Recovering the Optimal Policy

- **Once we obtain an optimal value functions, we can recover the corresponding optimal policy** (a plan of actions for the agent)
- Always take a better action that leads us to a better action value!
  - You can think of this as a one-step search. If you have the optimal value function then the actions that appear best after a one-step search will be optimal actions
  - Another way of saying this is that any policy that is greedy with respect to the optimal evaluation function is an optimal policy

Milestone Questions

- A Grid world
- You are given the optimal state value function
- Identify the best action(s) from each cell in the second top row

22.0	24.4	22.0	19.4	17.5
19.8	22.0	19.8	17.8	16.0
17.8	19.8	17.8	16.0	14.4
16.0	17.8	16.0	14.4	13.0
14.4	16.0	14.4	13.0	11.7

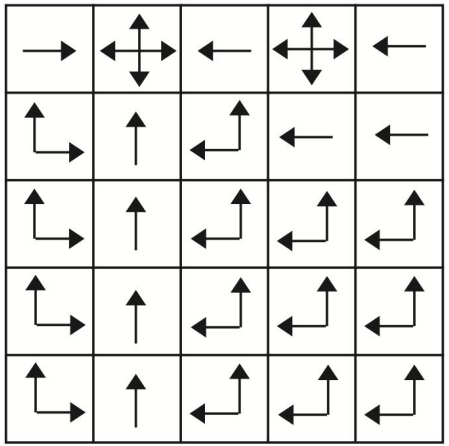
v_*

Milestone Questions

- A Grid world
- You are given the optimal state value function
- Identify the best action(s) from each cell in the second top row

22.0	24.4	22.0	19.4	17.5
19.8	22.0	19.8	17.8	16.0
17.8	19.8	17.8	16.0	14.4
16.0	17.8	16.0	14.4	13.0
14.4	16.0	14.4	13.0	11.7

v_*



π_*

Milestone Questions

- Can you tell the differences among reward, return, state-value function, and action-value function?
- Also, what's the difference between state values and "optimal" state values?

Milestone Questions

- Among the four concepts, which ones depend on a policy?
  - Reward
  - Return
  - state-value function
  - action-value function